

GenAI Ops: Indirect Prompt Injection Defense & Log Poisoning Firewall

Duration : 90 Min.

Scenario

Build a log pre-processor firewall script ('log_firewall.py') that scans production logs, detects embedded prompt injection vectors, and outputs a clean dataset.

Target File Location & Creation

File to Create/Update: student_workspace/log_firewall.py **Input Resource File to Inspect:** student_workspace/production_stream.log

Task Objectives

Perform the following actions inside the student_workspace directory: - Inspect production_stream.log in student_workspace/. - Create or update student_workspace/log_firewall.py and populate it with valid analysis results.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Open and inspect production_stream.log inside student_workspace/.
3. Create or open student_workspace/log_firewall.py in student_workspace/ and record your answers.
4. Save your changes (Ctrl + S or Cmd + S).
5. Open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.
6. Verify your progress by running python run.py locally in the terminal.

Validation

Once you have saved your files and verified your progress, return to the platform dashboard and click the **"Run Test" / "Verify"** button. This will automatically evaluate your changes and generate your score!

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	log_firewall.py is syntax-valid Python	3 Marks
TC2	Detects 'Ignore previous instructions' injection vector	3 Marks

TC3	Detects 'System:' / '[INSTRUCTION]' role hijacking attempts	3 Marks
TC4	Strips injection payloads while preserving real log lines	3 Marks
TC5	Outputs sanitized_logs.json	2 Marks
TC6	Generates security audit log entry for blocked injections	2 Marks
TC7	Calculates log sanitization statistics	2 Marks
TC8	Handles high-throughput log streams efficiently	2 Marks

Total Score: 20 Marks

Important Notes

- This is an auto-evaluated question. Ensure all code edits are properly saved and the 'run.py' checks pass before submission.

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